BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2021

PHYSICS

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

PHYSICS 9th — ANNUAL EXAM 2021 — SECTION –A

Time Allowed: 20 minutes | **Maximum Marks:** 12

Note: Q1: Choose the correct Answer.

Q1. Which prefix represents the largest value -------?

(A) Mega
(B) gaga
(C) peat
(D) exa

Q2. The average speed of bus is 20m/s. How Far can it travel in 10 seconds -------?

(A) 100m
(B) 150m
(C) 200m
(D) 250m

Q3. The formula to calculate the moment of force is ----

(A) Force /Area
(B) Force *velocity
(C) Force * Area
(D) None of these

Q4. The SI unit of inertia is -------.

(A) Newton
(B) meter per second
(C) meter
(D) kilogram

Q5. The SI unit of linear momentum is simply --------.

(A) Nm
(B) Ns
(C) N kg
(D) None of these

Q6. The shortest distance between two couple forces is....

(A) Momentum
(B) Couple arm
(C) radius
(D) diameter

Q7. Conventionally anti clockwise torque is taken as ----

(A) Negative
(B) positive
(C) parallel
(D) Zero

Q8. kWh is unit for ------.

(A) Energy
(B) power
(C) Force
(D) None of these

Q9. Young's Modulus for iron is -------.

(A) $11*10^{11}$ Pa
(B) $1.6*10^{10}$ Pa
(C) $21*10^{10}$ Pa
(D) $20*10^{10}$ Pa

Q10. Barometer is used to measure ------.

(A) Density
(B) vapor pressure
(C) 1kg * 1 meter
(D) None of these

Q11. One joule is equal to ----.

(A) 1meter x 1 sec
(B) 1 Newton x 1 meter
(C) 1kg x 1 meter
(D) None

Q12. Which of the following is the best heat conductor ---?

(A) Aluminum
(B) tin
(C) iron
(D) Copper

SECTION-B

Maximum Marks: 32

Q1. (Marks 32) Q1: Attempt any eight of the following questions. All carry equal marks.

i. Name any four derived units and write them as their base units?

ii. Differentiate between average and instantaneous velocities?

iii. Is it possible that displacement will be equal to distance?

iv. How can we feel inertia? Explain with an example.

v. What will be the effect if force is applied parallel to the exist of rotation?

vi. Moon is attracted by the earth, why it does not fall on earth?

vii. Can a centripetal force ever do work on an object? Explain.

viii. Differentiate between stress and strain?

ix. Why is water tanks constructed at the highest level in our houses?

x. The temperature of a normal human body is 37°C. Find this temperature on the Fahrenheit and Kelvin scales.

xi. Why wearing white is clothes preferred in summer?

SECTION- C

Maximum Marks: 21

Note: NOTE: Attempt any three of the following questions. All questions carry equal marks.

Q2.

i: Define scalar and vector quantities. Explain with examples the graphical representation of vector quantities.

ii: With what speed must a ball be thrown vertically from ground level to rise to a maximum height of 40m?

Q3.

i: Differentiate between centripetal force and centrifugal force.

ii: A ball of weight 100N is moving on a frictionless surface with a velocity of $10ms^{-1}$, compute its kinetic energy.

Q4.

i: State Pascal's principle and explain with examples.

ii: the deepest point in the ocean is 11 km below sea level, deeper than Mount Everest is tall. What is the pressure in atmosphere at this depth?

Q5.

i: What do you mean by thermal expansion? Prove that the coefficient of volume thermal expansion of solids 'γ' is about three times of the coefficient of Linear thermal expansion 'α' of solids.

ii: Consider a meter - stick composed of platinum for which a = $8.8 \times 10^{-6} k^{-1}$ by what amount does the length of this meter - stick change if the temperature is increased by 1 k?